

ERIC (CHEN-CHIEN) CHANG

Github: <https://github.com/unknowntpo/>

Email: e850506@gmail.com ◇ Phone: [+886-932786506](tel:+886-932786506)

Blog: <https://blog.unknowntpo.me/>

SUMMARY

- Backend / Data Infrastructure Engineer focused on **Kafka**, **PostgreSQL**, **distributed systems**, and **performance tuning**; delivered wins including first-byte latency **100s to 5s**, WebUI load time **60s to 1s**, and simulation jobs **2hr to 1min**.
- Core stack: **Go**, **Java**, **Rust**, **TypeScript**, **Kafka**, **Flink**, **Spark**, **ClickHouse**, **Iceberg**, **PostgreSQL**, **Airflow**, **Kubernetes**, **Docker**, **Cloudflare R2**.

OPEN SOURCE CONTRIBUTIONS

- **Apache Kafka Contributor**: collaborating on Kafka Streams **KIP-770** input-buffer byte bounds ([PR #22458](#)) and fixed offset commit-before-partition-switch correctness ([PR #21584](#)).
- **Apache Gravitino Committer**: added PostgreSQL/MySQL integration-test container management, reproducible Docker image workflows, and reduced Hive image size by **420MB**.
- **Apache DataFusion / Comet Contributor**: implemented Spark-compatible **str_to_map** in Rust and wired Comet's native SparkStrToMap UDF with SQL coverage.

SELECTED PROJECTS

TWFoundry

2026

*Real-time Taiwan transit data platform (hybrid cloud / homelab). Tech Stack: **Kafka**, **Flink**, **Iceberg**, **ClickHouse**, **Airflow**, **Kubernetes (k0s)**, **Cloudflare R2** / **Workers** / **Pages**, **Docker** [GitHub](#)*

- Built a real-time transit platform: **Kafka** ingestion into a partitioned **R2** lake on self-hosted **k0s Kubernetes**; a **Flink** speed layer for live anomaly detection (service-gap / bunching) and an **Airflow**-orchestrated **Iceberg** lakehouse + **ClickHouse** OLAP batch layer; edge-served via **Cloudflare Workers** / **Pages**.

EXPERIENCE

Lawson

January 2024 - December 2024

*Backend Software Engineer. Tech Stack: **Typescript**, **NodeJS**, **Docker**, **MongoDB**, **Prometheus**, **Grafana** Taipei, Taiwan*

- Implemented **HTTP streaming** for CPU-intensive computation results, reducing first-byte latency from **100s** to **5s**.
- Built a **MinIO events** and **Elasticsearch** monitoring path to detect missing data proactively and improve pipeline reliability.
- Built an **LLM service** for extracting information from ambiguous regulatory text and automated government act crawling, re-crawl, and re-parse workflows.

Mediatek (Contract)

February 2022 - March 2023

*Backend Software Engineer. Tech Stack: **Go**, **Linux**, **Docker**, **PostgreSQL** Hsinchu, Taiwan*

- Maintained a low-power simulation and benchmarking data warehouse, improving WebUI page load time from **60s** to **1s** with **PostgreSQL index-only scans**: [Blog Post](#).
- Used **Pyroscope eBPF** to identify import bottlenecks, then applied **PostgreSQL bulk import** to reduce simulation job duration from **2hr** to **1min**.

EDUCATION

National Taiwan Ocean University, Electrical Engineering coursework (Drop Out)

2013 - 2017